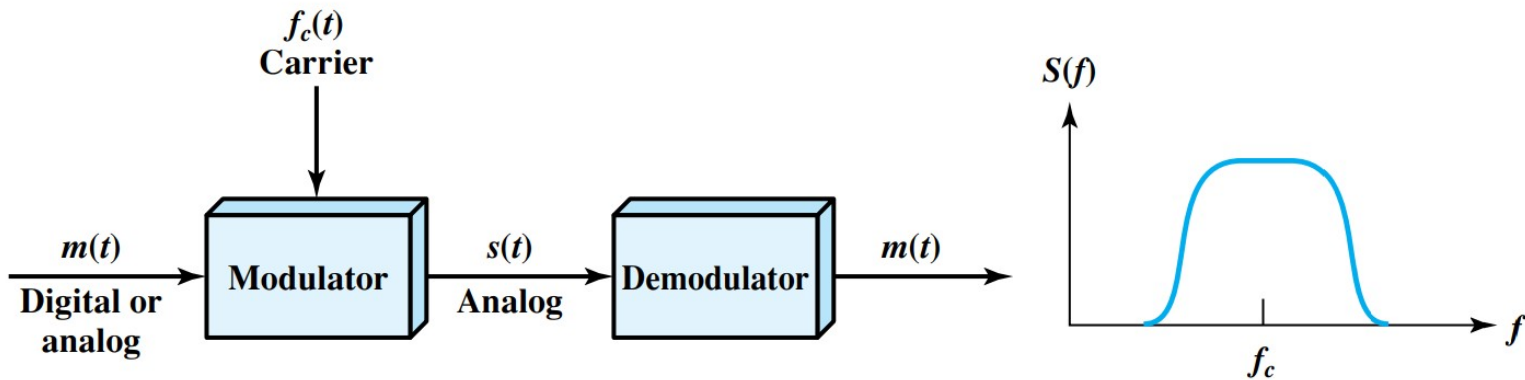


Computer Networks I

Signal Modulation Techniques (Analog to Analog)

Amitangshu Pal
Computer Science and Engineering
IIT Kanpur

Analog Data → Analog Signals



- Two principle techniques:
 - Amplitude modulation
 - Angle modulation
 - Frequency modulation
 - Phase modulation

Time Domain $\leftrightarrow$ Frequency

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

Linearity property: $x_1(t) \leftrightarrow X_1(f)$, $x_2(t) \leftrightarrow X_2(f) \equiv a_1 x_1(t) + a_2 x_2(t) \leftrightarrow a_1 X_1(f) + a_2 X_2(f)$

$$\begin{aligned} a_1 x_1(t) + a_2 x_2(t) &\leftrightarrow \int_{-\infty}^{\infty} \{a_1 x_1(t) + a_2 x_2(t)\} e^{-j2\pi ft} dt \\ &= \int_{-\infty}^{\infty} a_1 x_1(t) e^{-j2\pi ft} dt + \int_{-\infty}^{\infty} a_2 x_2(t) e^{-j2\pi ft} dt = a_1 X_1(f) + a_2 X_2(f) \end{aligned}$$

Time Domain $\Leftrightarrow$ Frequency

Domain

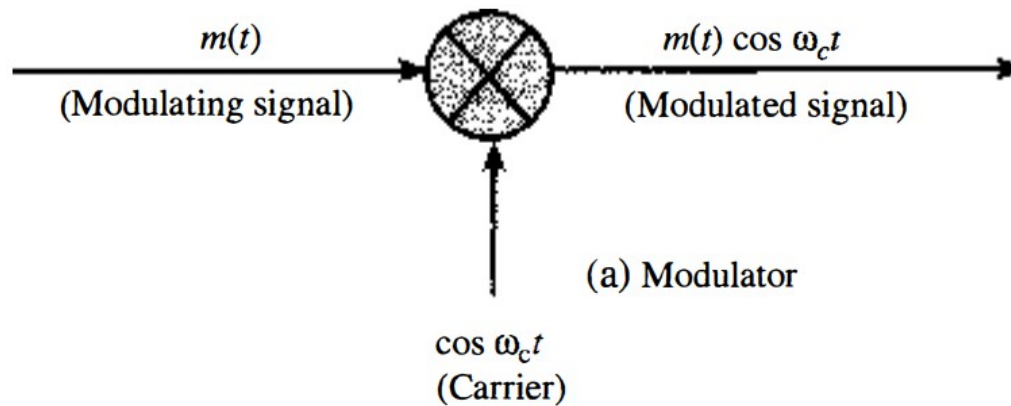
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

Frequency shifting property: $x(t) \leftrightarrow X(f) \equiv e^{j2\pi f_c t} x(t) \leftrightarrow X(f - f_c)$

$$e^{j2\pi f_c t} x(t) \leftrightarrow \int_{-\infty}^{\infty} e^{j2\pi f_c t} x(t) e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} x(t) e^{-j2\pi (f - f_c) t} dt = X(f - f_c)$$

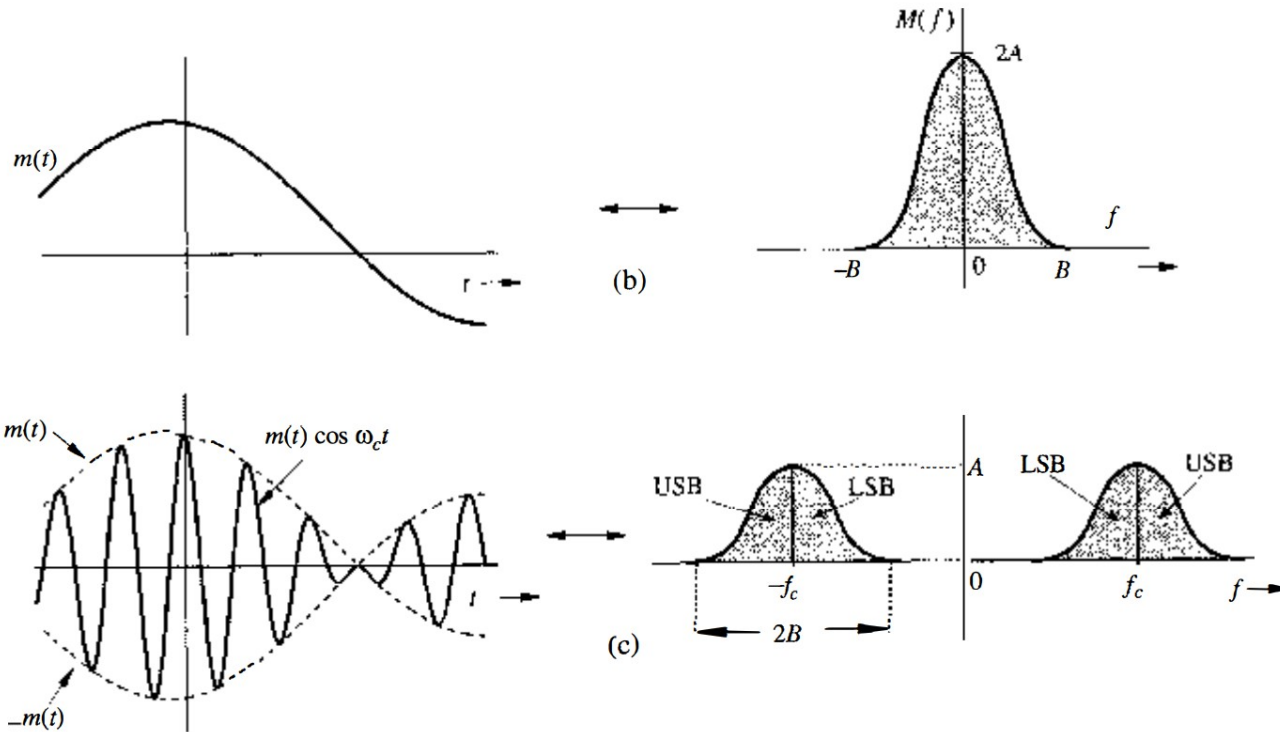
DSB-SC Modulation

Transmitted signal: $m(t) \cos(2\pi f_c t)$



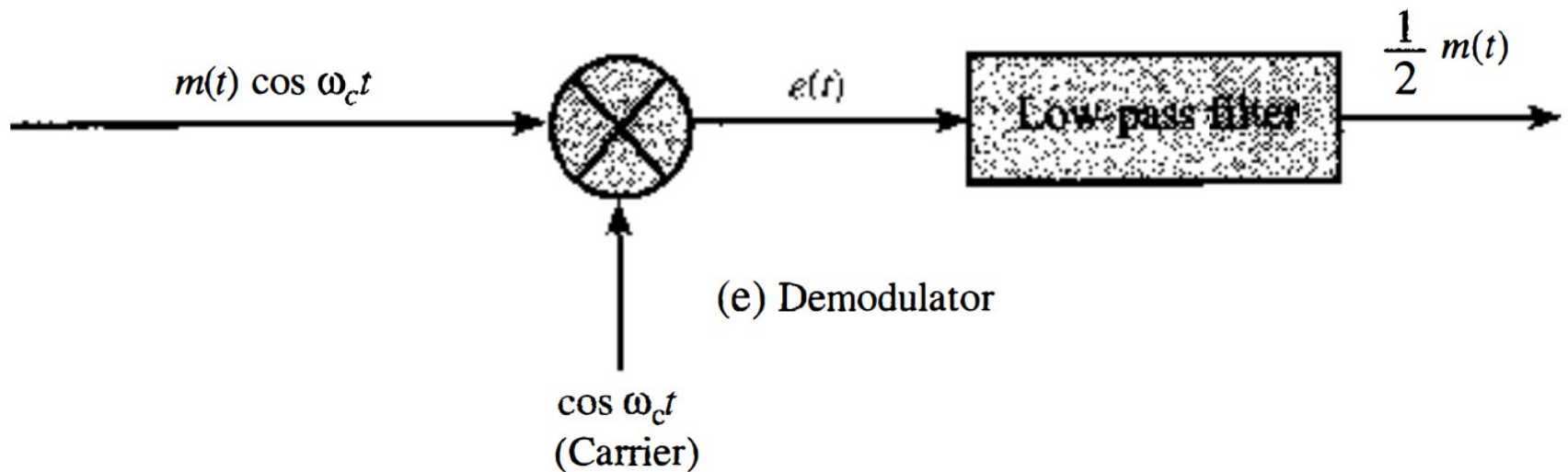
$$\begin{aligned}\mathbf{F}[m(t) \cos(2\pi f_c t)] &= \mathbf{F}\left\{\left(\frac{e^{j2\pi f_c t} + e^{-j2\pi f_c t}}{2}\right) m(t)\right\} \\ &= \frac{1}{2} [\mathbf{F}\{e^{j2\pi f_c t} m(t)\} + \mathbf{F}\{e^{-j2\pi f_c t} m(t)\}] = \frac{1}{2} [M(f - f_c) + M(f + f_c)]\end{aligned}$$

DSB-SC Modulation



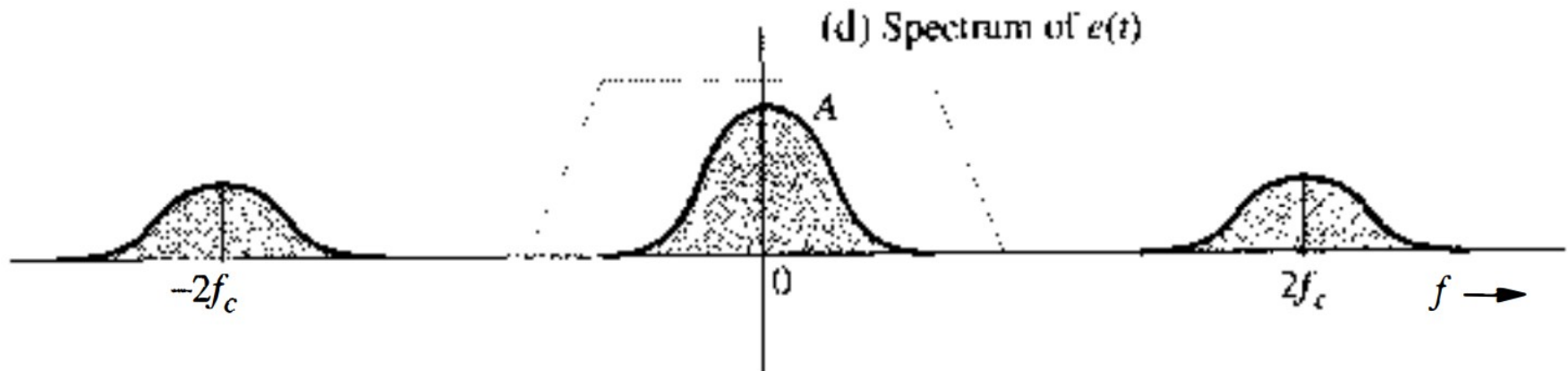
$$\mathbf{F}[m(t) \cos(2\pi f_c t)] = \frac{1}{2} [M(f - f_c) + M(f + f_c)]$$

DSB-SC Demodulation



$$\begin{aligned} \mathbf{F}[m(t) \cos^2(2\pi f_c t)] &= \mathbf{F}\left\{\left(\frac{1 + \cos(2\pi 2f_c t)}{2}\right) m(t)\right\} \\ &= \frac{1}{2} M(f) + \frac{1}{4} [M(f - 2f_c) + M(f + 2f_c)] \end{aligned}$$

DSB-SC Modulation



$$\begin{aligned}\mathbf{F}[m(t) \cos^2(2\pi f_c t)] &= \mathbf{F}\left\{\left(\frac{1 + \cos(2\pi 2f_c t)}{2}\right)m(t)\right\} \\ &= \frac{1}{2}M(f) + \frac{1}{4}[M(f - 2f_c) + M(f + 2f_c)]\end{aligned}$$

Angle Modulation

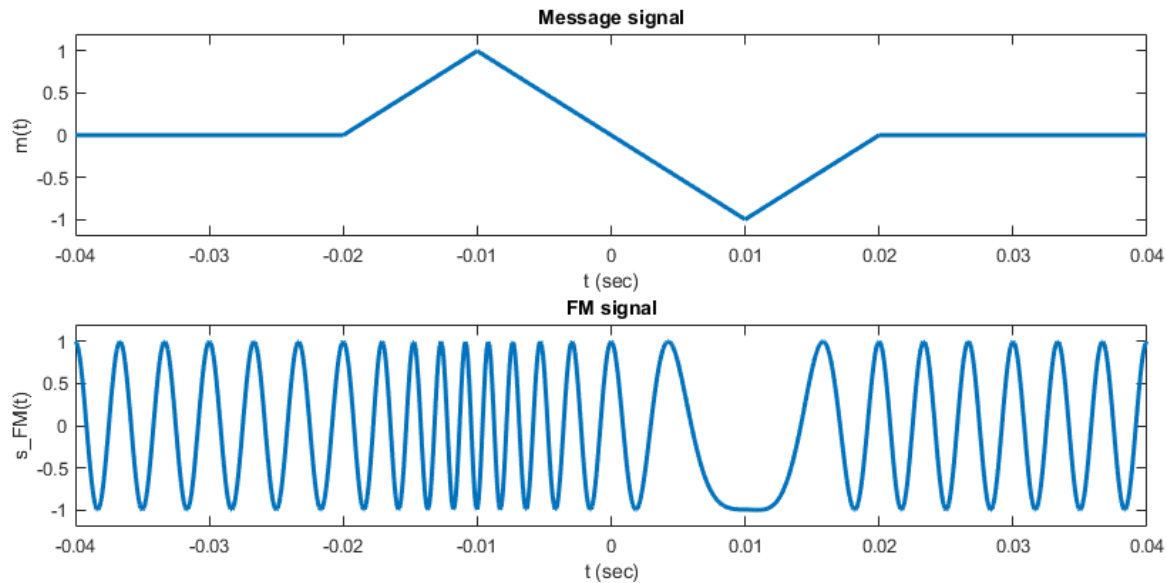
$$s(t) = A \cos \theta(t)$$

- Angle modulation:
 - Frequency modulation
 - Phase modulation
-

Angle Modulation (Frequency Modulation)

$$w_i(t) = 2\pi f_c + k_f m(t)$$

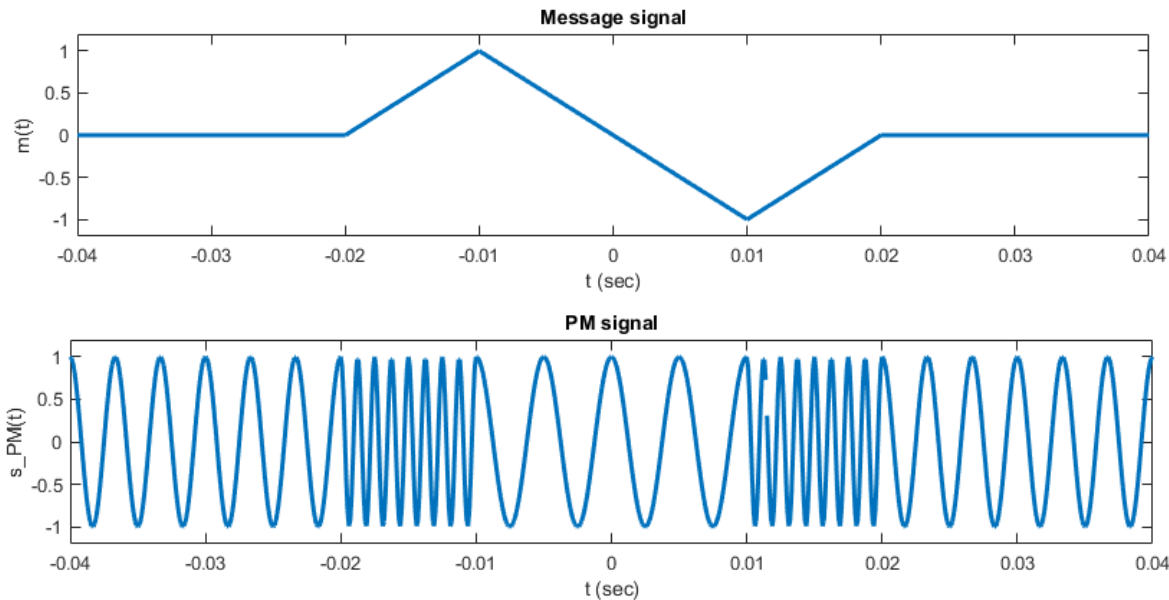
$$\therefore s_{\text{FM}}(t) = \cos((2\pi f_c + k_f m(t))t)$$



Angle Modulation (Phase Modulation)

$$\theta(t) = 2\pi f_c t + k_p m(t)$$

$$\therefore s_{PM}(t) = \cos(2\pi f_c t + k_p m(t))$$



AM vs FM/PM

□ Amplitude modulation:

- Bandwidth is less
- More susceptible to noise → noise affects the amplitude

□ Angle modulation:

- Modulated signal requires greater bandwidth than AM
 - More robust against noise, interference
-

Summary

□ Modulation techniques (Analog data → Analog signals):

- Different modulation techniques discussed
 - Amplitude modulation
 - Frequency modulation
 - Phase modulation
 - Comparison of the analog modulation techniques
-

Extras

DSB-TC Modulation

- ❑ The carrier is sent along with the message

Transmitted signal: $A \cos(2\pi f_c t) + m(t) \cos(2\pi f_c t)$

$$\begin{aligned} \mathbf{F}[A \cos(2\pi f_c t) + m(t) \cos(2\pi f_c t)] &= \mathbf{F}[A \cos(2\pi f_c t)] + \mathbf{F}[m(t) \cos(2\pi f_c t)] \\ &= \frac{A}{2} [\delta(f - f_c) + \delta(f + f_c)] + \frac{1}{2} [M(f - f_c) + M(f + f_c)] \end{aligned}$$

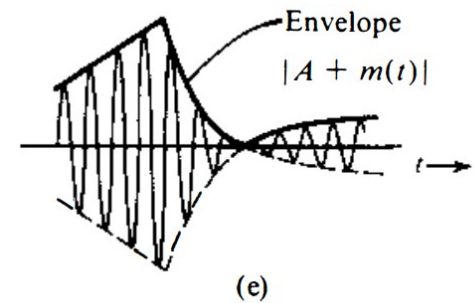
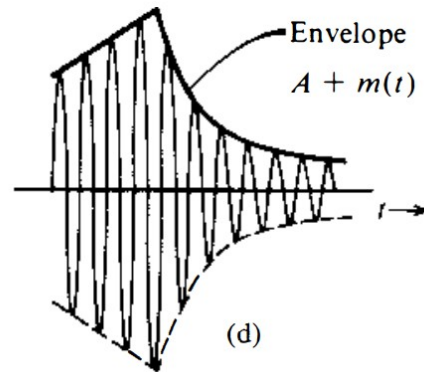
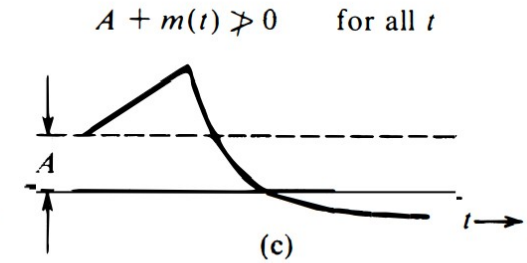
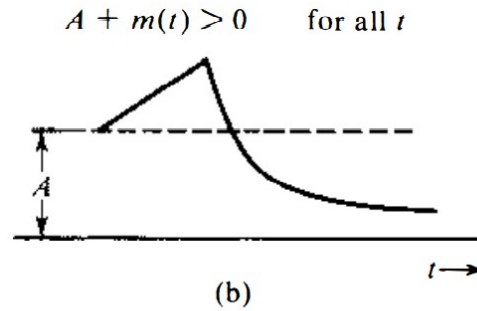
DSB-TC Modulation

$$A + m(t) \geq 0$$

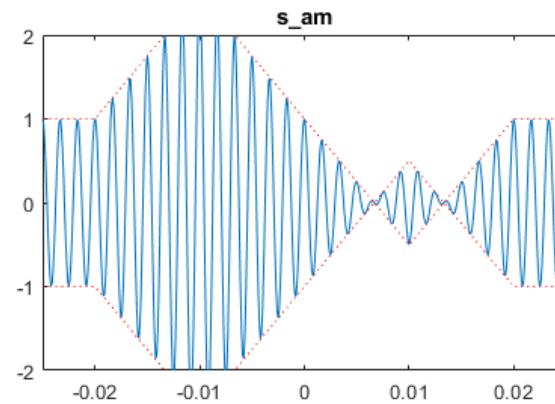
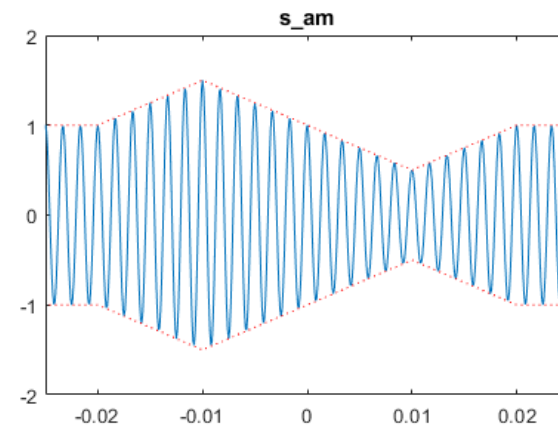
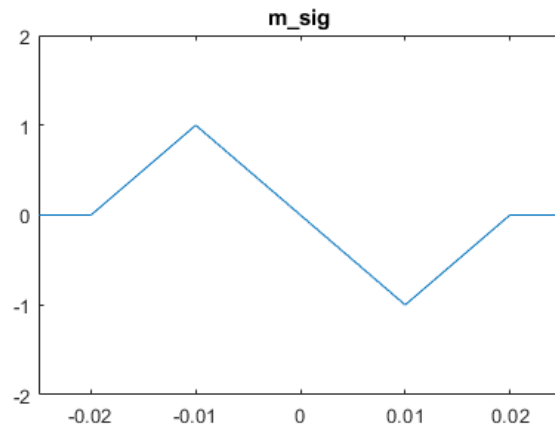
$$\therefore A \geq -m(t)$$

$$\text{Now } m(t) \geq -m_p \quad \therefore A \geq m_p$$

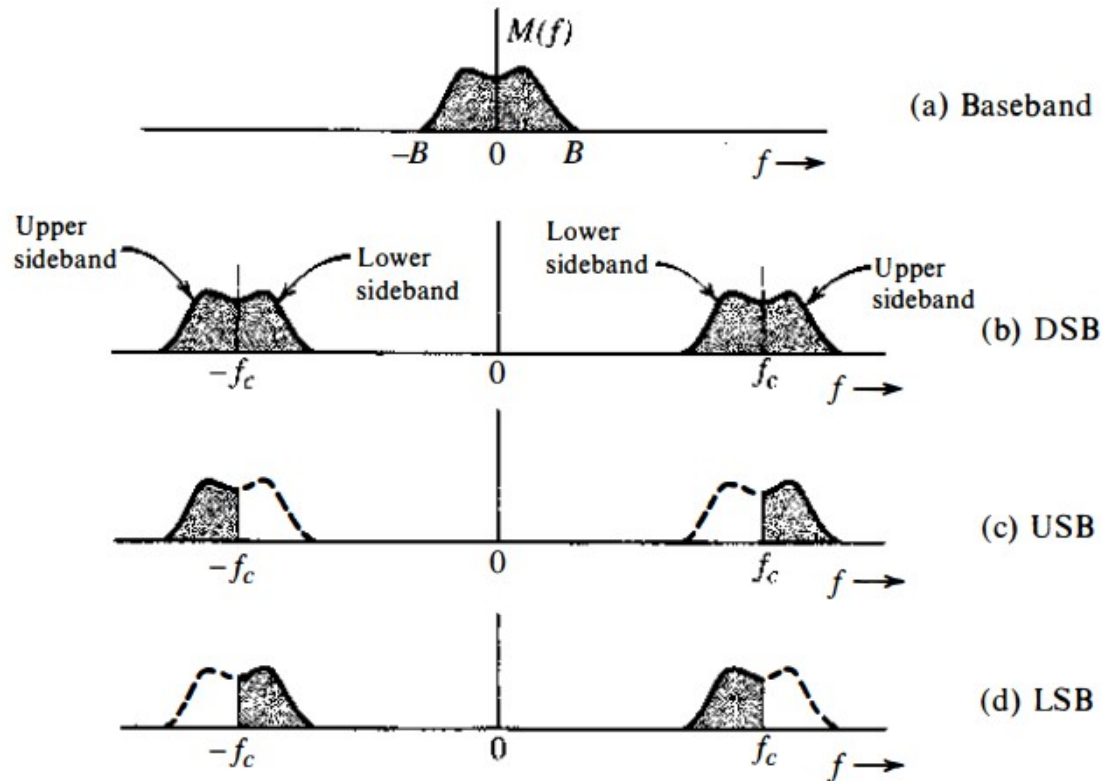
$$k_a = \frac{m_p}{A} \quad \therefore 0 \leq k_a \leq 1$$



DSB-TC Modulation



SSB Modulation



Angle Modulation

$$s(t) = A \cos \theta(t)$$

Instantaneous angular frequency is $\omega_i(t) = \frac{d\theta}{dt}$

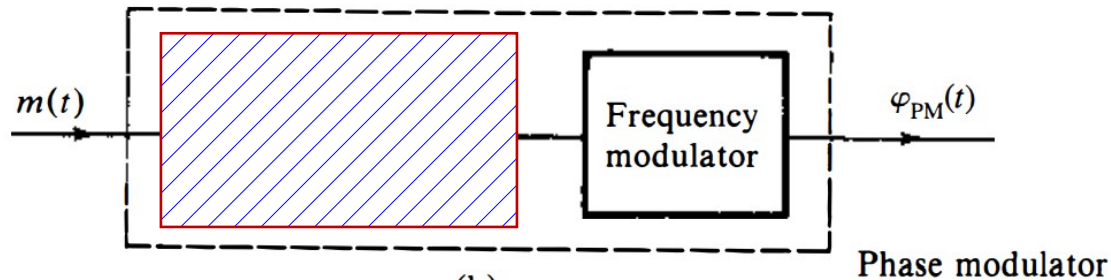
$$\omega_i(t) = 2\pi f_c + k_f m(t)$$

$$\therefore s_{\text{FM}}(t) = \cos((2\pi f_c + k_f m(t))t)$$

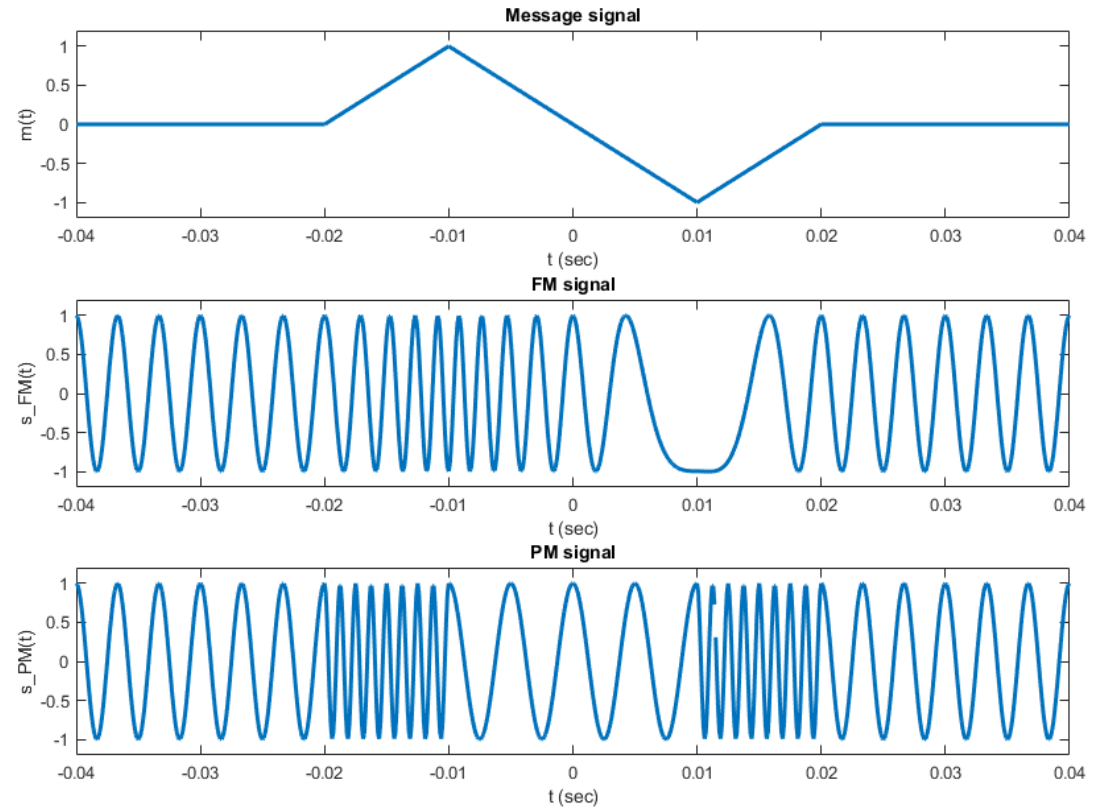
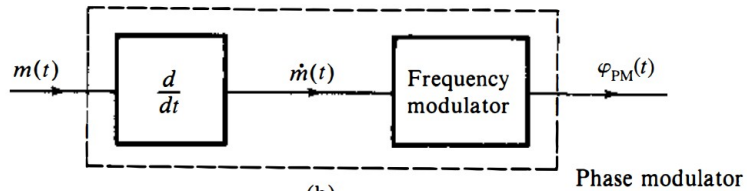
$$\theta(t) = 2\pi f_c t + k_p m(t)$$

$$\therefore s_{\text{PM}}(t) = \cos(2\pi f_c t + k_p m(t))$$

Instantaneous angular frequency is $\omega_i(t) = \frac{d\theta}{dt} = 2\pi f_c + k_p \dot{m}(t)$



Angle Modulation



Angle Modulation

